

# Hammad Amer

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Software Engineer at the intersection of Artificial Intelligence, Full-Stack Development, and Cloud. Deploys PyTorch models within MERN architectures and ships end-to-end products — from ML pipelines and DevSecOps automation to web and mobile apps.

## EDUCATION

**FAST NUCES** | *Islamabad*

**Aug 2022 - Jun 2026**

Bachelor of Science in Computer Science

**Coursework:** Data Structures & Algorithms, Generative AI, MLOps, Cloud Computing, Web Programming, Information Security

## TECHNOLOGIES

**Core Development:** React, Node.js, Express.js, Python, C++, Java, Kotlin, MongoDB, MySQL, Firebase

**Cloud & Infrastructure:** AWS, Microsoft Azure, Kubernetes, Docker, Terraform, GitHub Actions, ArgoCD, Linux

**AI & MLOps:** PyTorch, TensorFlow, LLMs, RAG, MLflow, Airflow

**Engineering Practices:** Agile, CI/CD, Test Driven Development (TDD)

## WORK EXPERIENCE

**Code Generation | AI/ML Developer Intern**

**Jun 2025 – Sep 2025**

- Architected and developed an automated Financial Data Parser leveraging multi-agent architectures and genetic algorithms, transitioning complex financial datasets into structured, actionable formats for the reconciliation system.
- Owned the end-to-end integration of these machine learning models into the startup's software applications, collaborating closely with the engineering team to ensure seamless deployment and reliable performance.
- Streamlined data processing workflows by delivering a production-ready parsing solution, demonstrating the ability to rapidly design and ship complex AI features within a fast-paced environment.

**Independent Software & Web Consultant**

**May 2023 – Feb 2024**

- Delivered freelance web and technical consulting for international clients, improving website performance and search visibility through infrastructure audits, technical SEO, and performance tuning.

## PROJECTS

**SpineScan (Final Year Project)** – AI Lumbar-Spine MRI Diagnostic Tool | **Stack:** PyTorch, YOLOX, timm, MERN

**GitHub:** github.com/Hammad-Amer/SpineScan

- Designed a multi-stage deep learning pipeline (YOLOX + timm backbones) classifying 25 lumbar-spine conditions from DICOM MRI scans at 94% multi-label accuracy.
- Cut training time 60% with FP16 mixed precision and gradient accumulation, and built a Grad-CAM explainability dashboard for single-study DICOM inference.

**Multimodal RAG & Semantic Search System** | **Stack:** LLaMA2, Sentence-BERT, CLIP/BLIP, FAISS, LlamaIndex

**GitHub:** github.com/Hammad-Amer/MultiModalRag

- Built a retrieval-augmented pipeline answering text and image queries from PDFs, using LLMs for context-aware responses with chain-of-thought and few-shot prompting.
- Engineered semantic retrieval over FAISS with CLIP/BLIP and Sentence-BERT embeddings, surfacing ranked chunks and images via a query interface.

**DevSecOps Netflix Pipeline** | **Stack:** Jenkins, Kubernetes (EKS), ArgoCD, AWS, SonarQube, Trivy

**GitHub:** github.com/Hammad-Amer/devsecops-netflix-pipeline

- Built an end-to-end DevSecOps pipeline that builds, security-scans, and deploys a containerized app to Amazon EKS, with Jenkins CI/CD gated by SonarQube, Trivy, and OWASP.
- Implemented GitOps continuous delivery with ArgoCD and full Prometheus/Grafana observability.

## ACHIEVEMENTS & HONORS

**Dean's List Honoree (2×)** — FAST NUCES Islamabad